MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE 0BH

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.4 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be DBH.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/159-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 *Circuit pin configuration.* The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The modules shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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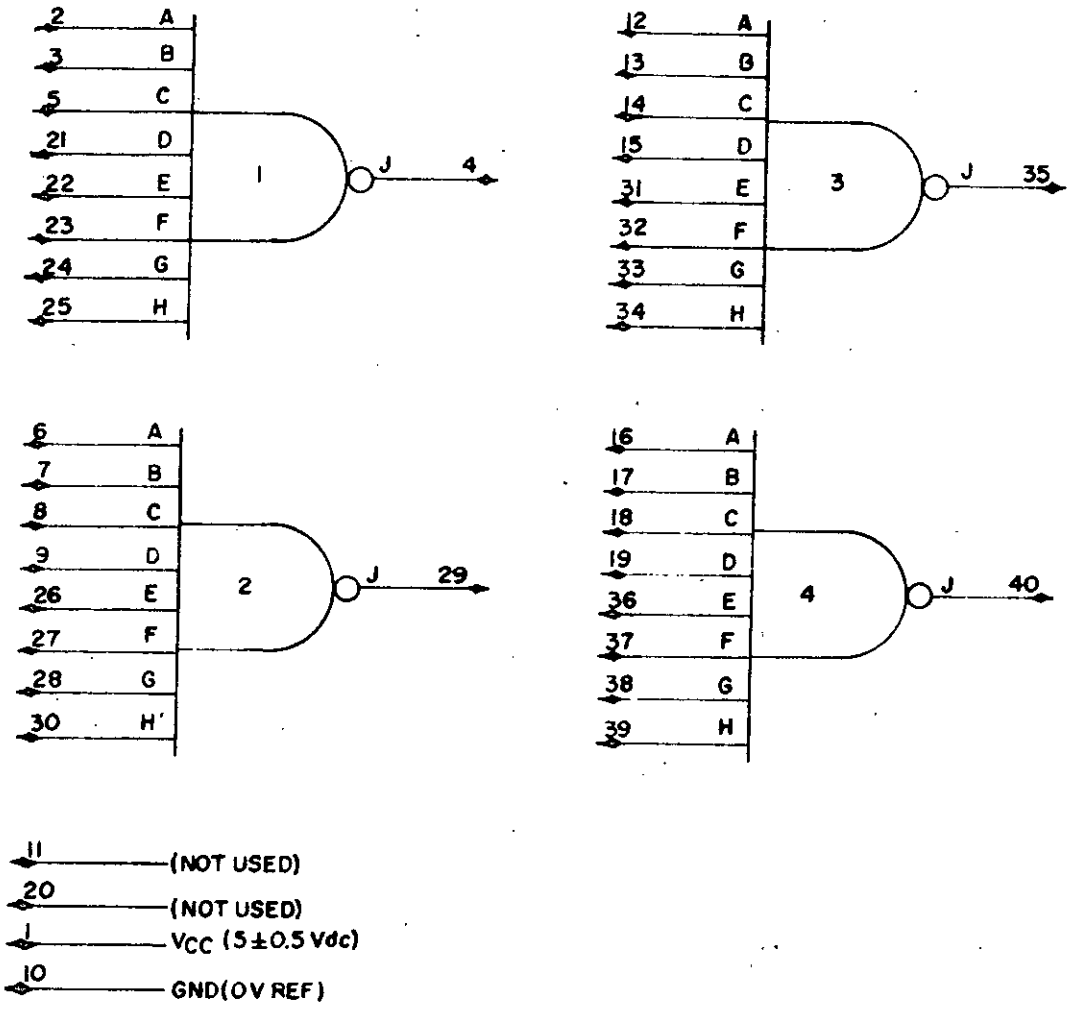


FIGURE 1. DBH module function diagram, NAND gate.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				0° to 60°C Operating temperature				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A4	V_{OUT} (low-loaded)	III	S	0.0	0.6	0.0	0.6	0.0	0.6	0.0	0.6	V
B1-B32	Input breakdown current	IV	S	0	100	0	100	0	100	0	100	µA
C1	Isolation, frame	V	S	10	∞	1	∞	10	∞	1	∞	MΩ
C2-C3	Isolation, unused pins	V	X	10	∞	1	∞	10	∞	1	∞	MΩ
D1-D4	Output voltage, low level	VI	X	0.0	0.3	0.0	0.3	0.0	0.3	0.0	0.3	V
E1-E32	Output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F1-F32	Input current, low level	VIII	X	50	200	50	200	50	200	50	200	µA
G1-G32	Input current, high level	IX	X	0	10	0	10	0	10	0	10	µA
H1-H4	Output short circuit current	X	X	3	15	3	15	3	15	3	15	mA
J1	Power supply current, low level	XI	S	0.2	2.0	0.2	2.0	0.2	2.0	0.2	2.0	mA
J2	Power supply current, high level	XI	S	0.2	1.3	0.2	1.3	0.2	1.3	0.2	1.3	mA
K1-K4	Transition time (t_{TLH}), C_L = 40 pF	XII	S	5	60	5	60	5	60	5	60	ns
K5-K8	Transition time (t_{TLH}), C_L = 140 pF	XII	Q	10	100	10	100	10	100	10	100	ns
L1-L4	Transition time (t_{THL}), C_L = 40 pF	XIII	S	2	30	2	30	2	30	2	30	ns
L5-L8	Transition time (t_{THL}), C_L = 140 pF	XIII	Q	5	60	5	60	5	60	5	60	ns
M1-M4	Propagation delay time (t_{PLH}), C_L = 40 pF	XIV	S	10	60	10	60	10	70	10	70	ns
M5-M8	Propagation delay time (t_{PLH}), C_L = 140 pF	XIV	Q	10	90	10	90	10	100	10	100	ns
N1-N4	Propagation delay time (t_{PHL}), C_L = 40 pF	XV	S	10	110	10	110	10	115	10	115	ns
N5-N8	Propagation delay time (t_{PHL}), C_L = 140 pF	XV	Q	10	130	10	130	10	145	10	145	ns
P1-P4	Output crosstalk, low level	XVI	Q	0	400	0	400	0	400	0	400	mV
R1-R4	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	mV
S	Power supply transient amplitude	XVIII	Q	0	300	0	300	0	300	0	300	mV

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				0° to 60°C Operating temperature				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
T	Power supply filter capacitance	XIX	S	1.65	2.75	1.65	2.75	1.65	2.75	1.65	2.75	µF
U1-U32	Input capacitance	XX	Q	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent, or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through T shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through T shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through T shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/159-1 supersedes part number M28787/159. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	
All outputs at low level	12 mW (outputs not loaded)
All outputs at high level	8 mW (outputs not loaded)
Supply voltage	5 Vdc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.7 V maximum
High level	2.0 V minimum, V_{CC} maximum

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Input current

Low level	(V_{IL} = 0.0 V)	-0.2 mA maximum
	(V_{IL} = 0.3 V)	-0.18 mA maximum
High level	(V_{IH} = 2.4 V)	10 µA maximum
	(V_{IH} = 5.5 V)	100 µA maximum

Output voltage

Low level (I_{OL} = 2.0 mA)	0.3 V maximum
High level (I_{OH} = -100 µA)	2.4 V maximum

Output drive current maximum capability

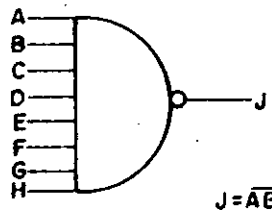
Low level (V_{OL} = 0.3 V)	2.0 mA
High level (V_{OH} = 2.4 V)	-100 µA

6.5 Module function. This module contains four independent low power 8-input NAND gates.

6.5.1 Input to output relationship. Each NAND gate has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.

INPUT	OUTPUT
AT LEAST ONE INPUT IN LOW STATE	HIGH STATE
ALL INPUTS IN HIGH STATE	LOW STATE



6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

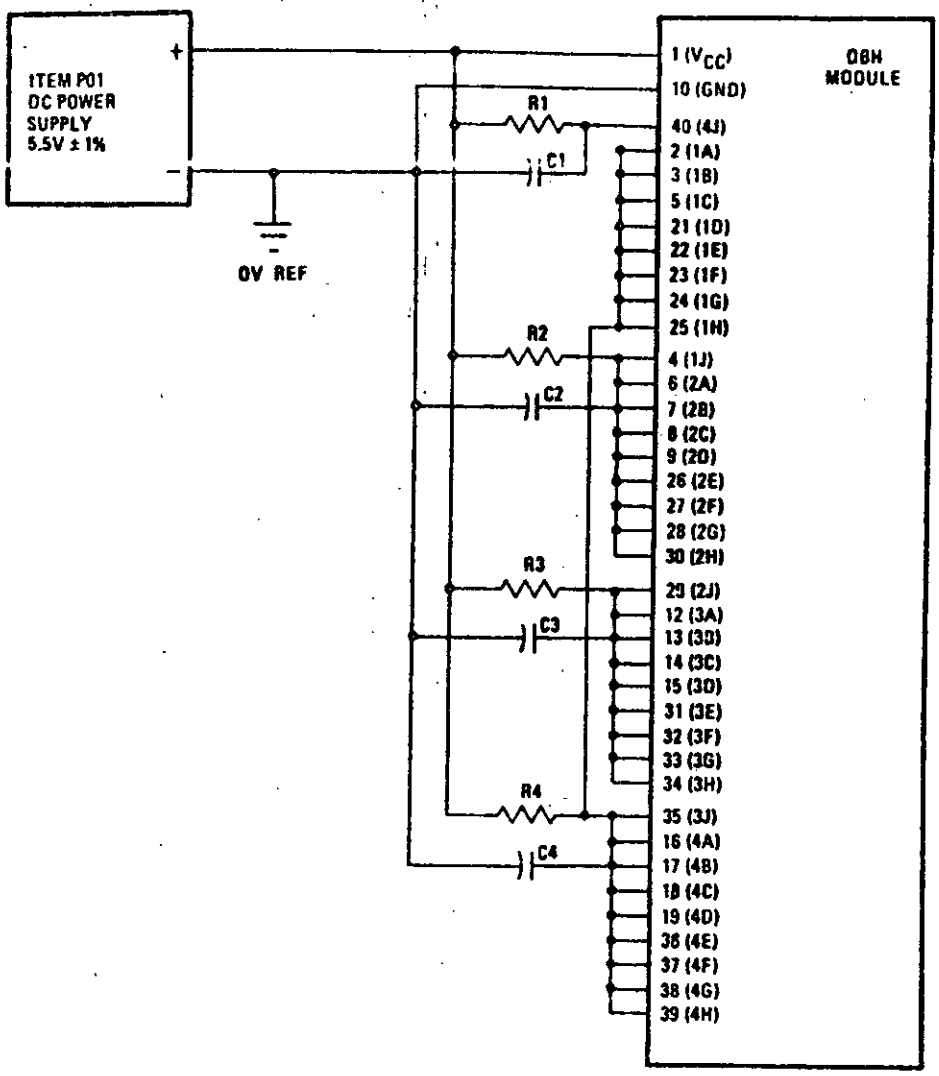
6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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NOTES:
1. R1-R4 = 2.75 kΩ ±1%, 1/8 W.
2. C1-C4 = 100 pF ±20%.

FIGURE 2. Operating temperature and life test circuit.

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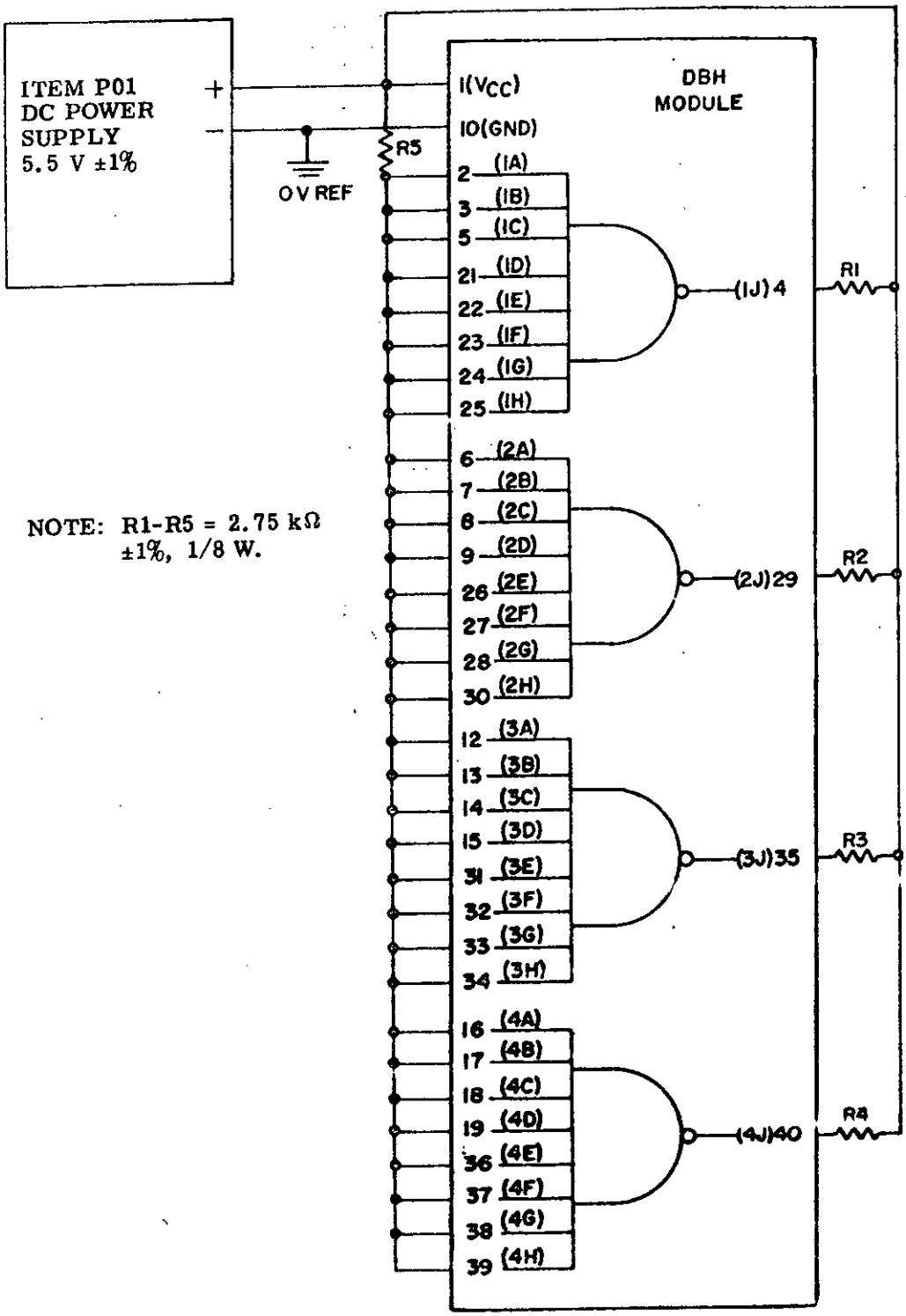


FIGURE 3. Component temperature test circuit.

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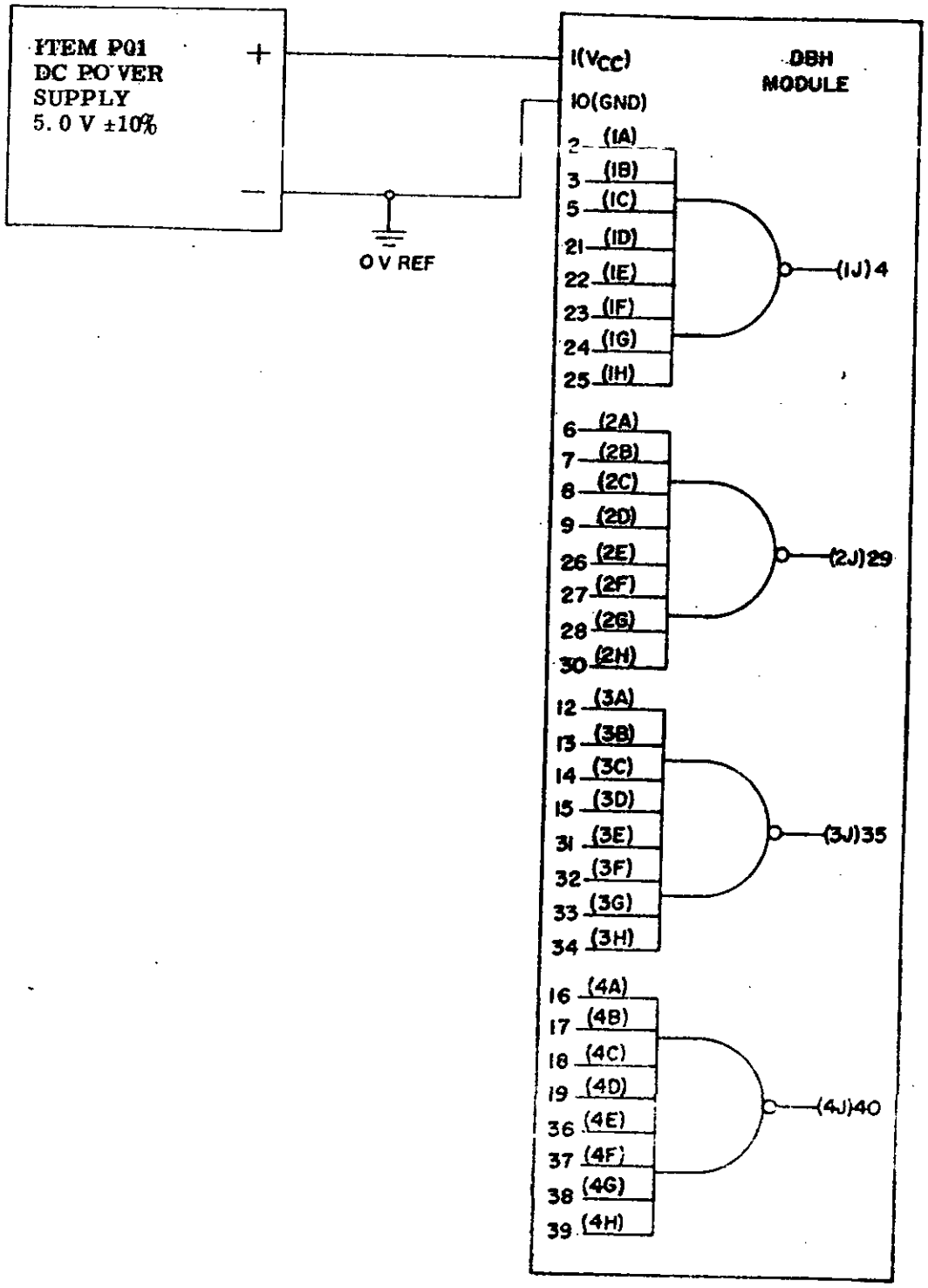


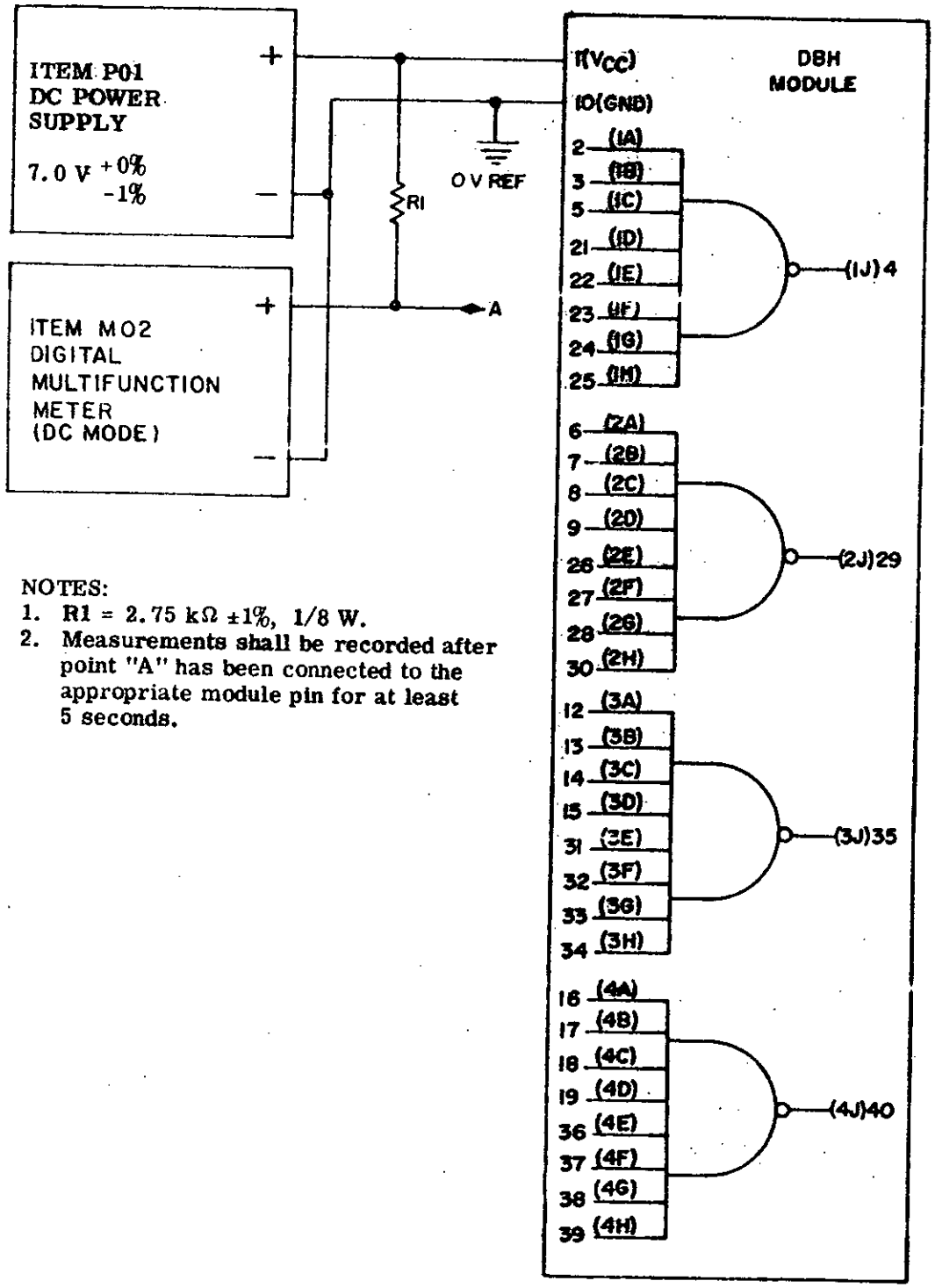
FIGURE 4. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
A1	1J V_{OUT} (low loaded)	4	0.0	0.6	V	5
A2	2J	29	↓	↓	↓	↓
A3	3J	35				
A4	4J ↓	40				

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NOTES:
1. R1 = 2.75 kΩ ±1%, 1/8 W.
2. Measurements shall be recorded after point "A" has been connected to the appropriate module pin for at least 5 seconds.

FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1A Input breakdown current	2	3,5,21,22,23,24,25	0	100	µA	6
B2	1B	3	2,5,21,22,23,24,25				
B3	1C	5	2,3,21,22,23,24,25				
B4	1D	21	2,3,5,22,23,24,25				
B5	1E	22	2,3,5,21,23,24,25				
B6	1F	23	2,3,5,21,22,24,25				
B7	1G	24	2,3,5,21,22,23,25				
B8	1H	25	2,3,5,21,22,23,24				
B9	2A	6	7,8,9,26,27,28,30				
B10	2B	7	6,8,9,26,27,28,30				
B11	2C	8	6,7,9,26,27,28,30				
B12	2D	9	6,7,8,26,27,28,30				
B13	2E	26	6,7,8,9,27,28,30				
B14	2F	27	6,7,8,9,26,28,30				
B15	2G	28	6,7,8,9,26,27,30				
B16	2H	30	6,7,8,9,26,27,28				
B17	3A	12	13,14,15,31,32,33,34				
B18	3B	13	12,14,15,31,32,33,34				
B19	3C	14	12,13,15,31,32,33,34				
B20	3D	15	12,13,14,31,32,33,34				
B21	3E	31	12,13,14,15,32,33,34				
B22	3F	32	12,13,14,15,31,33,34				
B23	3G	33	12,13,14,15,31,32,34				

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TABLE IV. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B24	3H Input breakdown current	34	12,13,14,15,31,32,33	0	100	μA	6
B25	4A	16	17,18,19,36,37,38,39	↓	↓	↓	↓
B26	4B	17	16,18,19,36,37,38,39				
B27	4C	18	16,17,19,36,37,38,39				
B28	4D	19	16,17,18,36,37,38,39				
B29	4E	36	16,17,18,19,37,38,39				
B30	4F	37	16,17,18,19,36,38,39				
B31	4G	38	16,17,18,19,36,37,39				
B32	4H	39	16,17,18,19,36,37,38				

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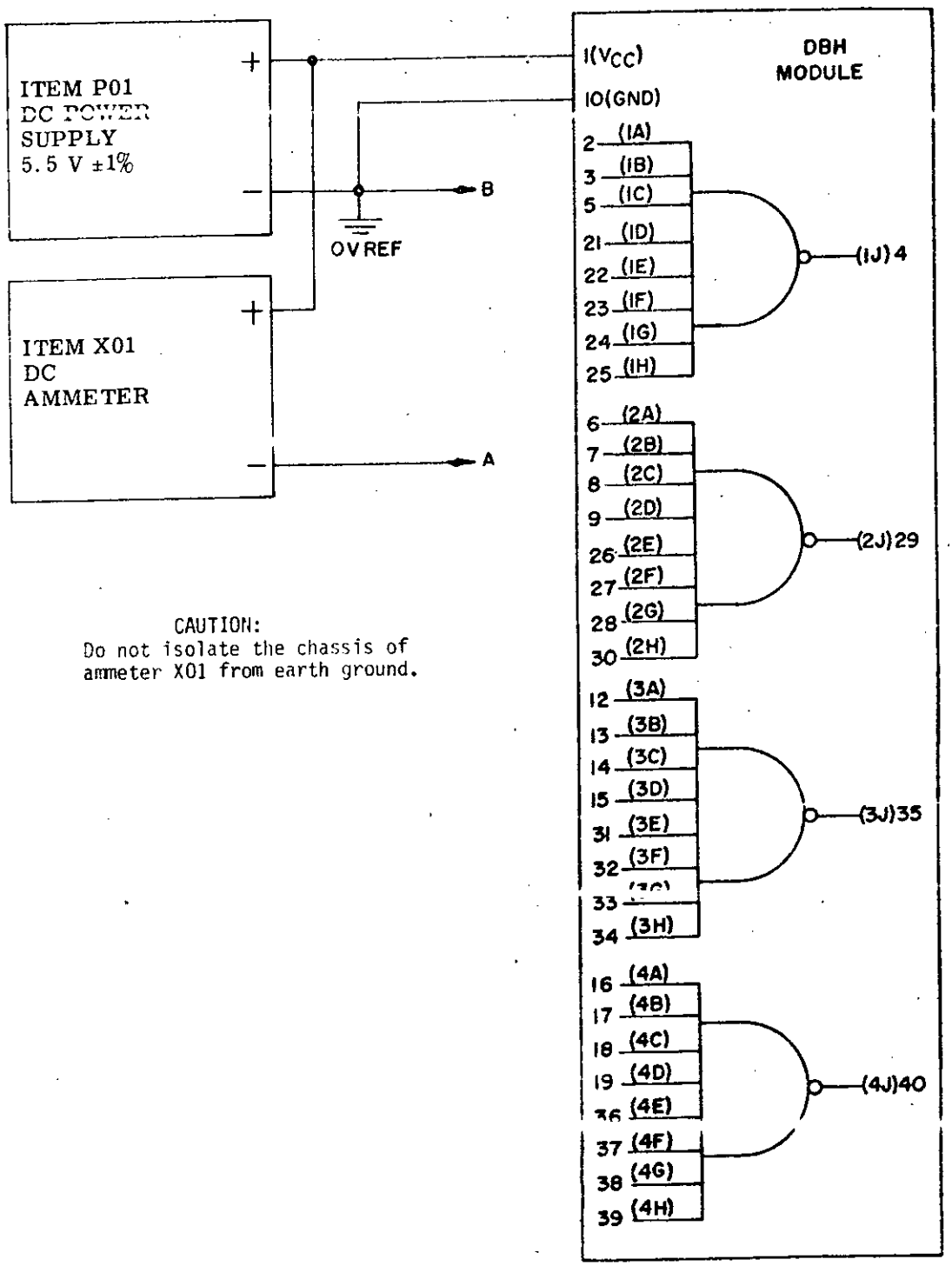


FIGURE 6. Input breakdown current test circuit for table IV.

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TABLE V. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Isolation, frame to module pins	C, 11, 20	Frame	10	∞	MΩ	7
C2	Isolation, used to unused pins	C	11, 20	10	∞	MΩ	7
C3	Isolation, unused to unused pins	11	20	10	∞	MΩ	7

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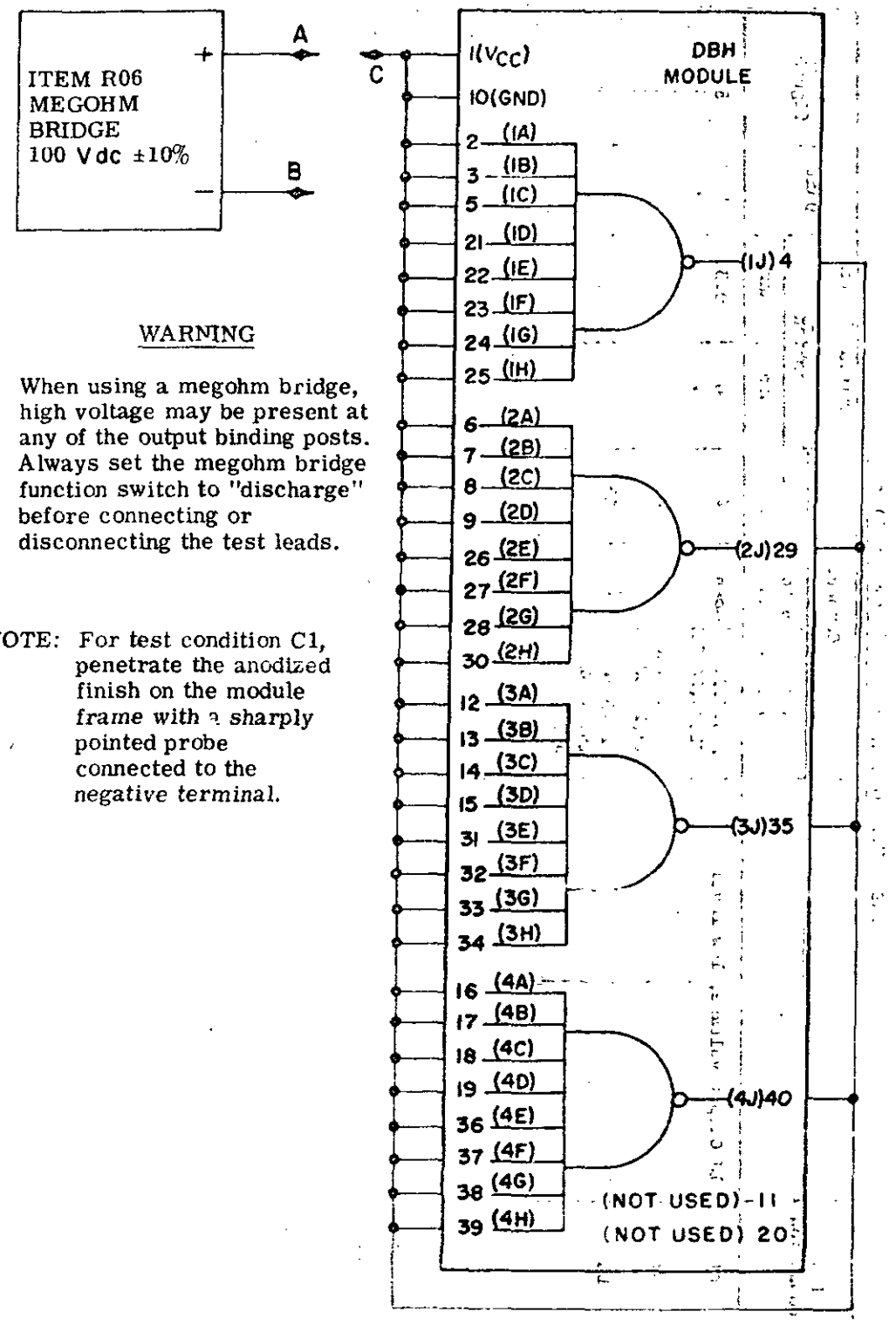


FIGURE 7. Isolation test circuit for table V.

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TABLE VI. Output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	1J Output voltage, low level	2,3,5,21, 22,23,24,25	Open	4	0	0.3	V	8
D2	2J	6,7,8,9,26, 27,28,30		29				
D3	3J	12,13,14, 15,31,32, 33,34		35				
D4	4J	16,17,18, 19,36,37, 38,39		40				

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TABLE VII. Output voltage (high-level) test.

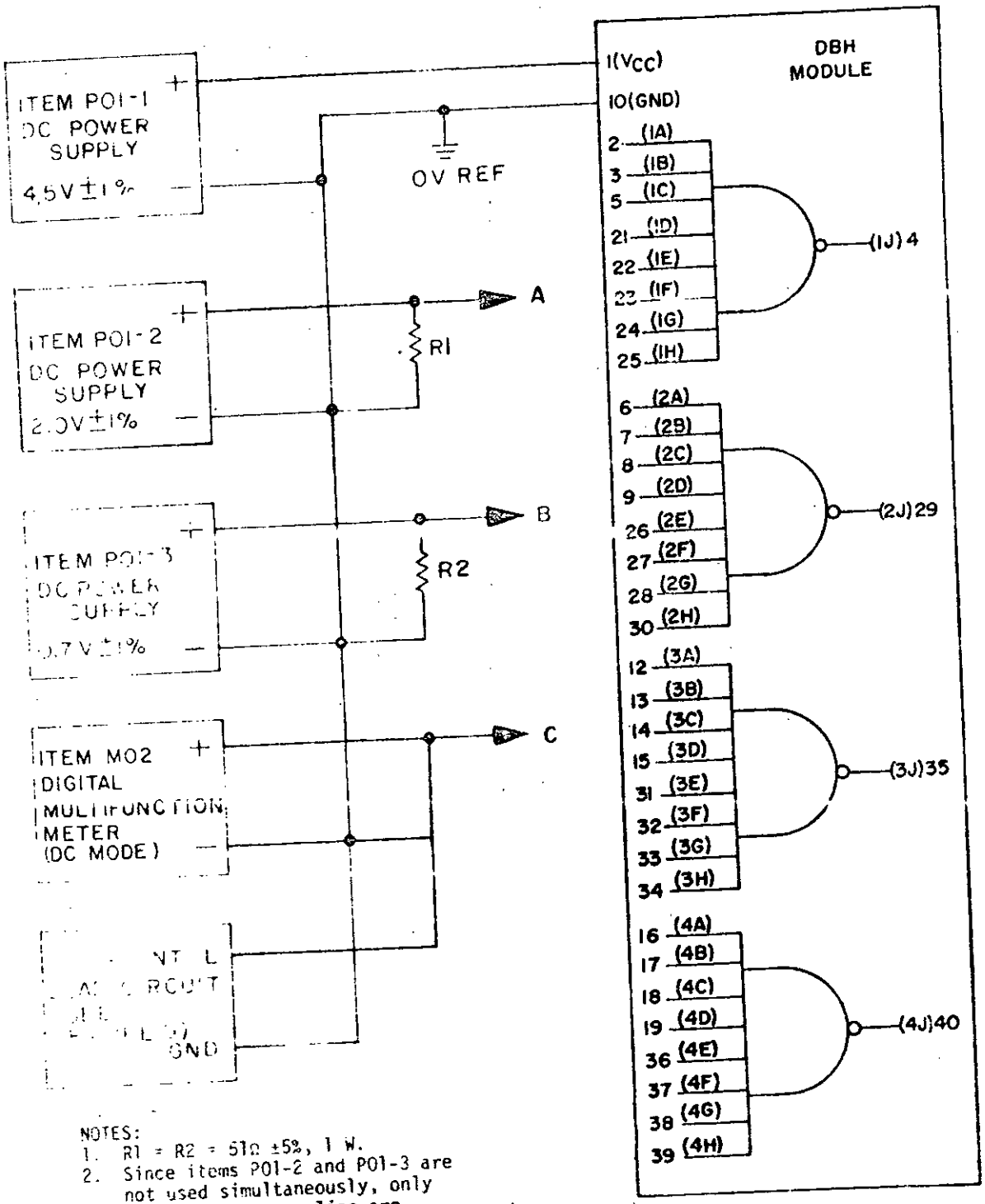
Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	1J. Output voltage, high level	Open	2	4	2.4	4.5	V	8
E2	1J		3					
E3	1J		5					
E4	1J		21					
E5	1J		22					
E6	1J		23					
E7	1J		24					
E8	1J		25					
E9	2J		6	29				
E10	2J		7					
E11	2J		8					
E12	2J		9					
E13	2J		26					
E14	2J		27					
E15	2J		28					
E16	2J		30					
E17	3J		12	35				
E18	3J		13					
E19	3J		14					
E20	3J		15					
E21	3J		31					
E22	3J		32					
E23	3J		33					
E24	3J		34					

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TABLE VII. Output voltage (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E25	4J Output voltage, high level	Open	16	40	2.4	4.5	V	8
E26	4J		17					
E27	4J		18					
E28	4J		19					
E29	4J		36					
E30	4J		37					
E31	4J		38					
E32	4J		39					

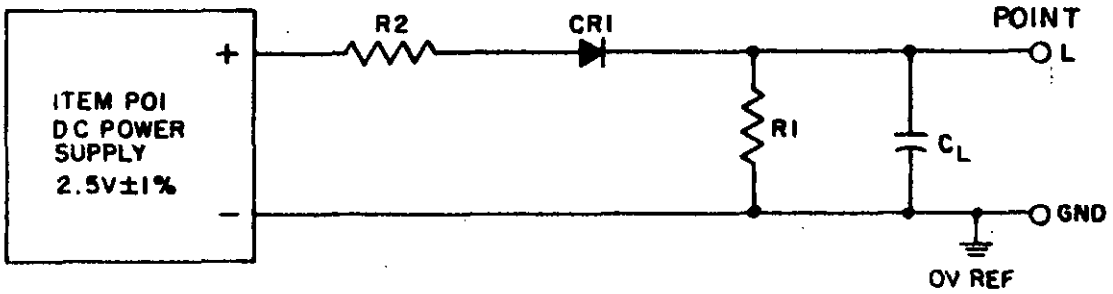
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Since items PO1-2 and PO1-3 are not used simultaneously, only two PO1 power supplies are required.

FIGURE 8. Output voltage (low-level and high-level) test circuit for tables VI and VII.

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NOTES:
1. CR1 = 1N916, or equivalent.
2. R1 = 23.7 kΩ ±1%, 1/8 W with a current flow of 100 µA when point L is held at 2.4 V.
3. R2 = Select value for current flow of 2.0 mA ±5% out of point L when point L is held at 0.3 V.
4. C_L = 40 ±5 pF (including probe capacitance) for test conditions K1-K4, L1-L4, M1-M4 and N1-N4.
   = 140 ±5 pF (including probe capacitance) for test conditions K5-K8, L5-L8, M5-M8 and N5-N8.
   = 50 ±5 pF (including probe capacitance) for test conditions P1-P4 and R1-R4.
5. C_L is not required for dc electrical tests D and E (table II).
6. The maximum lead length from point "L" to the appropriate module pin shall be less than 1 inch (25.4 mm).
7. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

FIGURE 9. Logic load circuit.

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TABLE VIII. Input current (low-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	1A Input current, low level	Open	2	Open	3,5,21,22, 23,24,25	50	200	μA	10
F2	1B		3		2,5,21,22, 23,24,25				
F3	1C		5		2,3,21,22, 23,24,25				
F4	1D		21		2,3,5,22, 23,24,25				
F5	1E		22		2,3,5,21, 23,24,25				
F6	1F		23		2,3,5,21, 22,24,25				
F7	1G		24		2,3,5,21, 22,23,25				
F8	1H		25		2,3,5,21, 22,23,24				
F9	2A		6		7,8,9,26, 27,28,30				
F10	2B		7		6,8,9,26, 27,28,30				
F11	2C		8		6,7,9,26, 27,28,30				
F12	2D		9		6,7,8,26, 27,28,30				
F13	2E		26		6,7,8,9, 27,28,30				
F14	2F		27		6,7,8,9, 26,28,30				

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TABLE VIII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F15	2G Input current, low level	Open	28	Open	6,7,8,9, 26,27,30	50	200	µA	10
F16	2H		30		6,7,8,9, 26,27,28				
F17	3A		12		13,14,15, 31,32,33,34				
F18	3B		13		12,14,15, 31,32,33,34				
F19	3C		14		12,13,15, 31,32,33,34				
F20	3D		15		12,13,14, 31,32,33,34				
F21	3E		31		12,13,14, 15,32,33,34				
F22	3F		32		12,13,14, 15,31,33,34				
F23	3G		33		12,13,14, 15,31,32,34				
F24	3H		34		12,13,14, 15,31,32,33				
F25	4A		16		17,18,19, 36,37,38,39				
F26	4B		17		16,18,19, 36,37,38,39				
F27	4C		18		16,17,19, 36,37,38,39				
F28	4D		19		16,17,18, 36,37,38,39				
F29									
F30									
F31									

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TABLE VIII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F29	4E Input current, low level	Open	36	Open	16,17,18, 19,37,38,39	50	200	mA	10
F30	4F		37		16,17,18, 19,36,38,39				
F31	4G		38		16,17,18, 19,36,37,39				
F32	4H		39		16,17,18, 19,36,37,38				

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TABLE IX. Input current (high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	1A Input current, high level	2	Open	3,5,21,22, 23,24,25	Open	0	10	µA	10
G2	1B	3		2,5,21,22, 23,24,25					
G3	1C	5		2,3,21,22, 23,24,25					
G4	1D	21		2,3,5,22, 23,24,25					
G5	1E	22		2,3,5,21, 23,24,25					
G6	1F	23		2,3,5,21, 22,24,25					
G7	1G	24		2,3,5,21, 22,23,25					
G8	1H	25		2,3,5,21, 22,23,24					
G9	2A	6		7,8,9,26, 27,28,30					
G10	2B	7		6,8,9,26, 27,28,30					
G11	2C	8		6,7,9,26, 27,28,30					
G12	2D	9		6,7,8,26, 27,28,30					
G13	2E	26		6,7,8,9, 27,28,30					
G14	2F	27		6,7,8,9, 26,28,30					

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TABLE IX. Input current (high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G15	2G Input current, high level	28	Open	6,7,8,9, 26,27,30	Open	0	10	µA	10
G16	2H	30		6,7,8,9, 26,27,28					
G17	3A	12		13,14,15, 31,32,33,34					
G18	3B	13		12,14,15, 31,32,33,34					
G19	3C	14		12,13,15, 31,32,33,34					
G20	3D	15		12,13,14, 31,32,33,34					
G21	3E	31		12,13,14, 15,32,33,34					
G22	3F	32		12,13,14, 15,31,33,34					
G23	3G	33		12,13,14, 15,31,32,34					
G24	3H	34		12,13,14, 15,31,32,33					
G25	4A	16		17,18,19, 36,37,38,39					
G26	4B	17		16,18,19, 36,37,38,39					
G27	4C	18		16,17,19, 36,37,38,39					
G28	4D	19		16,17,18, 36,37,38,39					

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TABLE IX. Input current (high-level) test - Continued.

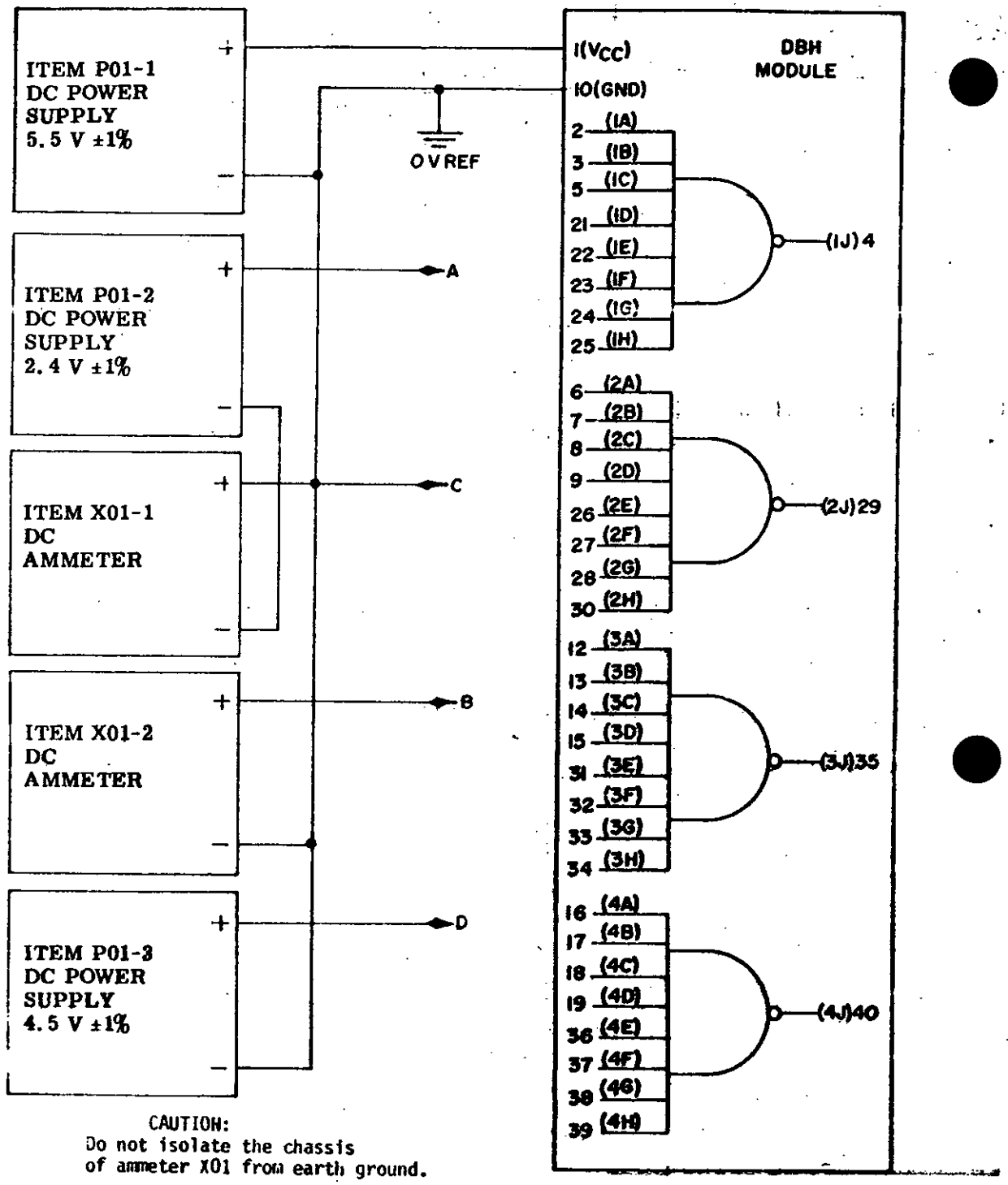
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G29	4E Input current, high level	36	Open	16,17,18, 19,37,38,39	Open	0	10	µA	10
G30	4F	37		16,17,18, 19,36,38,39					
G31	4G	38		16,17,18, 19,36,37,39					
G32	4H	39		16,17,18, 19,36,37,38					

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TABLE X. Output short circuit current test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	1J Output short circuit current	Open	4	21,22,23, 24,25,2,3,5	Open	3	15	mA	10
H2	2J		29	26,27,28, 30,6,7,8,9					
H3	3J		35	31,32,33,34, 12,13,14,15					
H4	4J		40	36,37,38,39, 16,17,18,19					

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CAUTION:
Do not isolate the chassis
of ammeter X01 from earth ground.

NOTES:
1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01, is required.
2. Since items P01-2 and P01-3 are not used simultaneously, only two item P01 power supplies are required.
3. With point "A" disconnected, item X01-1 should indicate zero current on the 1 µA range.
4. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.

FIGURE 10. Input current (low-level and high-level) and output short circuit current test circuit for tables VIII, IX and X.

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TABLE XI. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	Power supply current, low level	C	Open	A	0.2	2.0	mA	11
J2	Power supply current, high level	B	A	Open	0.2	1.3	mA	11

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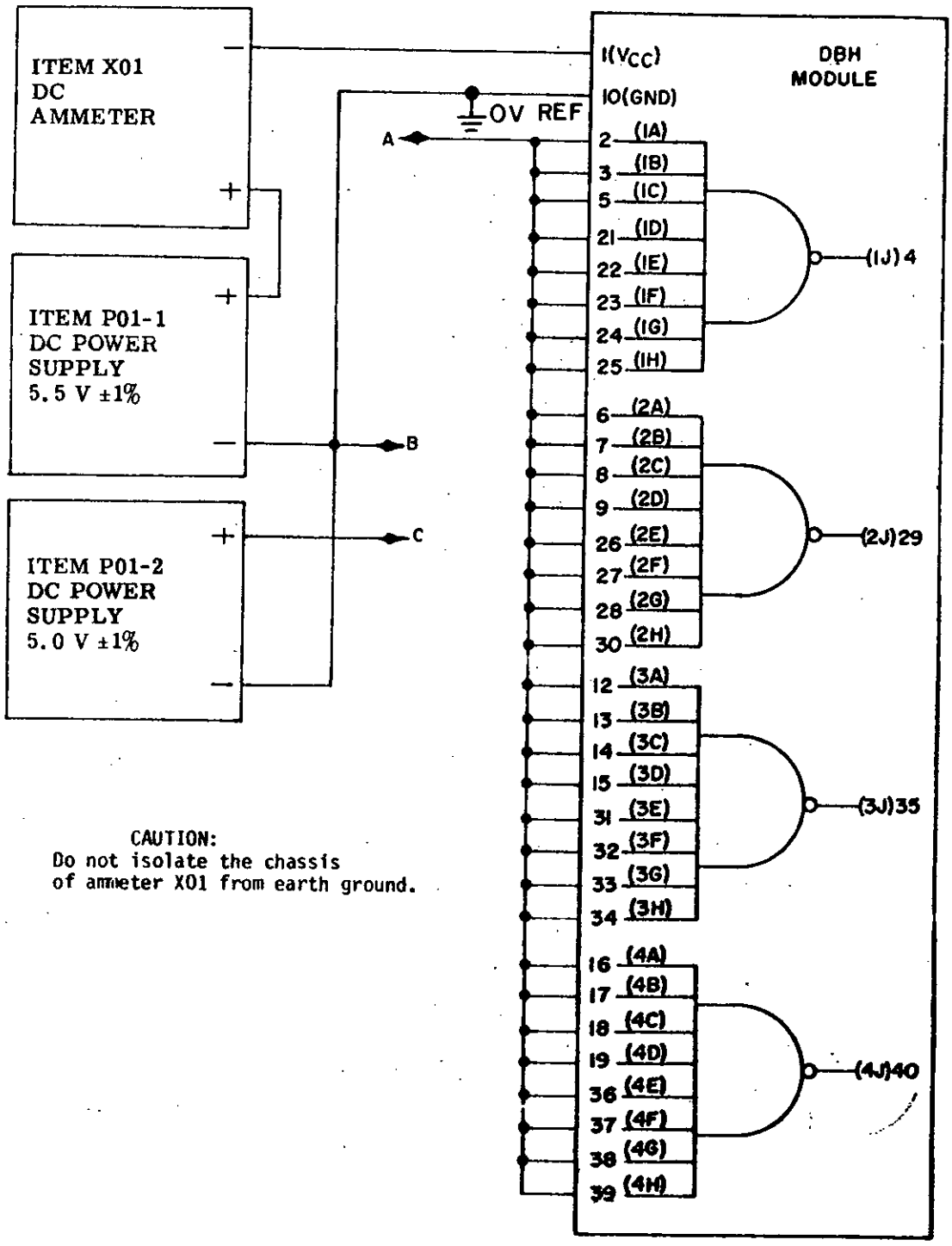


FIGURE 11. Power supply current (low-level and high-level) test circuit for table XI.

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TABLE XII. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	1J Transition time (t_{TLH}), C_L = 40 pF	2	3,5,21,22, 23,24,25	4	5	60	ns	12, 13
K2	2J	6	7,8,9,26, 27,28,30	29				
K3	3J	12	13,14,15,31, 32,33,34	35				
K4	4J	16	17,18,19,36, 37,38,39	40				
K5	1J Transition time (t_{TLH}), C_L = 140 pF	2	3,5,21,22, 23,24,25	4	10	100		
K6	2J	6	7,8,9,26, 27,28,30	29				
K7	3J	12	13,14,15,31, 32,33,34	35				
K8	4J	16	17,18,19,36, 37,38,39	40				

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TABLE XIII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	1J Transition time (t_{THL}), C_L = 40 pF	2	3,5,21,22, 23,24,25	4	2	30	ns	12, 13
L2	2J	6	7,8,9,26, 27,28,30	29	↓	↓	↓	↓
L3	3J	12	13,14,15,31, 32,33,34	35				
L4	4J	16	17,18,19,36, 37,38,39	40				
L5	1J Transition time (t_{THL}), C_L = 140 pF	2	3,5,21,22, 23,24,25	4	5	60		
L6	2J	6	7,8,9,26, 27,28,30	29	↓	↓		
L7	3J	12	13,14,15,31, 32,33,34	35				
L8	4J	16	17,18,19,36, 37,38,39	40				

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TABLE XIV. Propagation delay time (t_{PLH}) test.

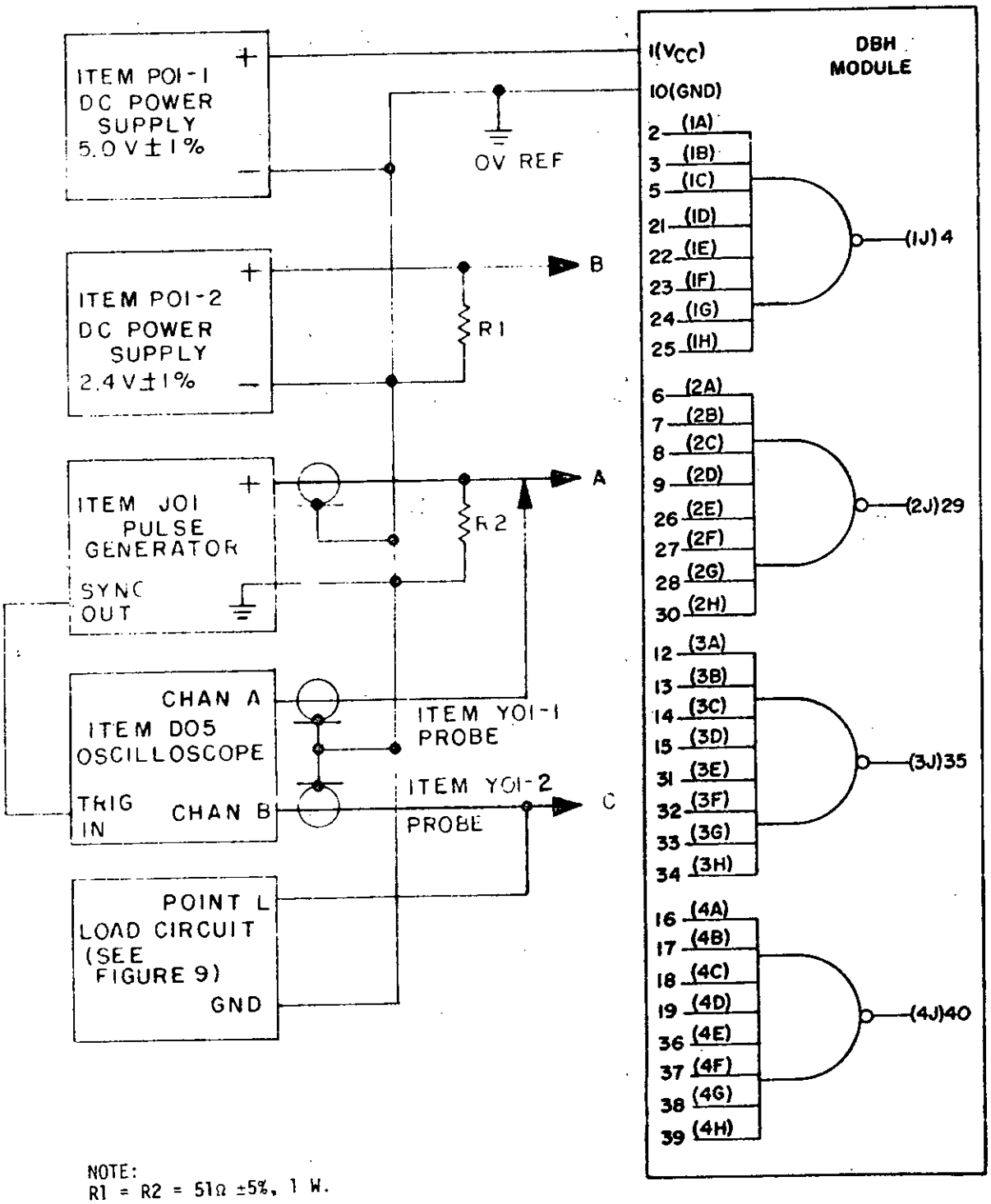
Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	1J Propagation delay time (t_{PLH}), C_L = 40 pF	2	3,5,21,22, 23,24,25	4	10	60	ns	12, 13
M2	2J	6	7,8,9,26, 27,28,30	29	↓	↓	↓	↓
M3	3J	12	13,14,15,31, 32,33,34	35				
M4	4J	16	17,18,19,36, 37,38,39	40				
M5	1J Propagation delay time (t_{PLH}), C_L = 140 pF	2	3,5,21,22, 23,24,25	4	10	90		
M6	2J	6	7,8,9,26,27, 28,30	29	↓	↓		
M7	3J	12	13,14,15,31, 32,33,34	35				
M8	4J	16	17,18,19,36, 37,38,39	40				

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TABLE XV. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	1J Propagation delay time (t_{PHL}), C_L = 40 pF	2	3,5,21,22 23,24,25	4	10	110	ns	12, 13
N2	2J	6	7,8,9,26, 27,28,30	29				
N3	3J	12	13,14,15, 31,32,33,34	35				
N4	4J	16	17,18,19, 36,37,38,39	40				
N5	1J Propagation delay time (t_{PHL}), C_L = 140 pF	2	3,5,21,22, 23,24,25	4		130		
N6	2J	6	7,8,9,26, 27,28,30	29				
N7	3J	12	13,14,15, 31,32,33,34	35				
N8	4J	16	17,18,19, 36,37,38,39	40				

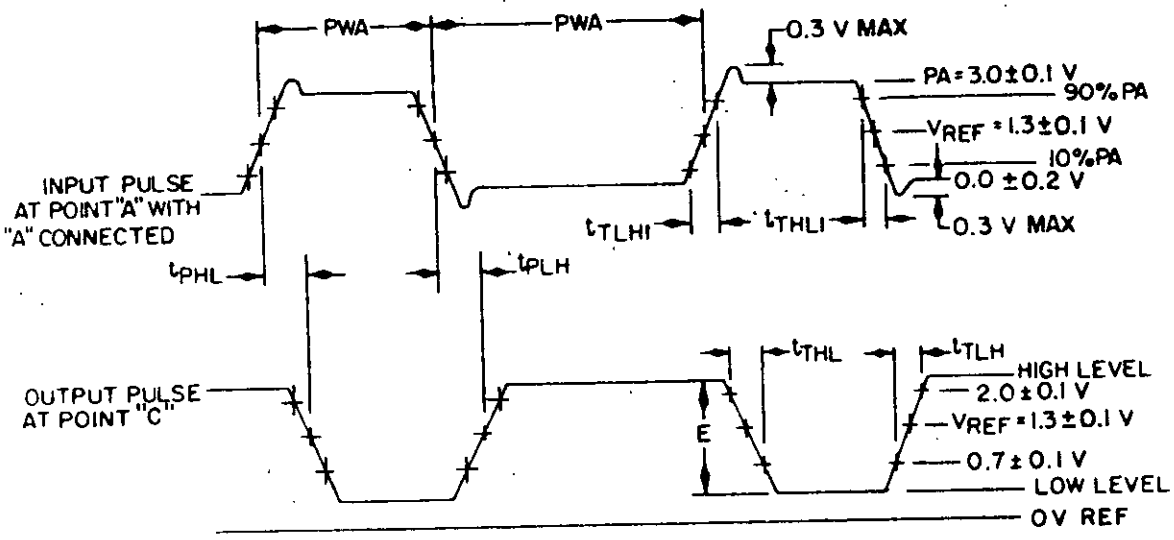
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NOTE:
R1 = R2 = 51Ω ±5%, 1 W.

FIGURE 12. Transition time, (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XII, XIII, XIV, and XV.

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NOTES:
1. Input pulse parameters:
   $t_{TLHI} = t_{THLI} = 10 \pm 2$ ns.
   PWA = 500 ±50 ns.
2. Before making the measurements verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.3 V.
3. All voltages are referenced to ground.

FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XII, XIII, XIV, and XV.

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TABLE XVI. Output crosstalk (low-level) test.

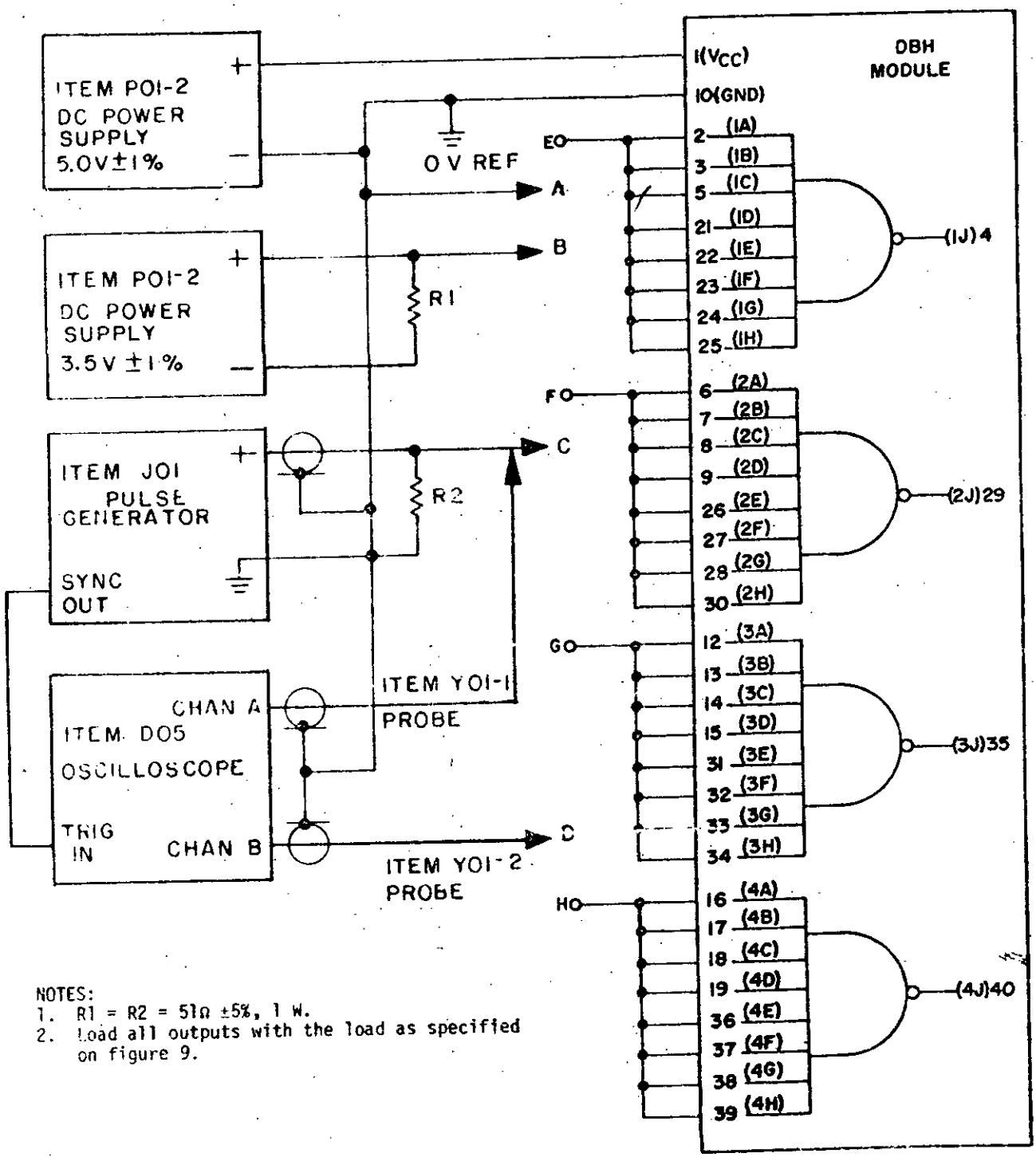
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1J Output crosstalk, low level	Open	E	F,G,H	4	0	400	mV	14, 15
P2	2J	↓	F	E,G,H	29	↓	↓	↓	↓
P3	3J		G	E,F,H	35				
P4	4J		H	E,F,G	40				

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TABLE XVII. Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1J Output crosstalk, high level	E	Open	F,G,H	4	0	400	mV	14, 15
R2	2J	F	↓	E,G,H	29	↓	↓	↓	↓
R3	3J	G		E,F,H	35				
R4	4J ↓	H		E,F,G	40				

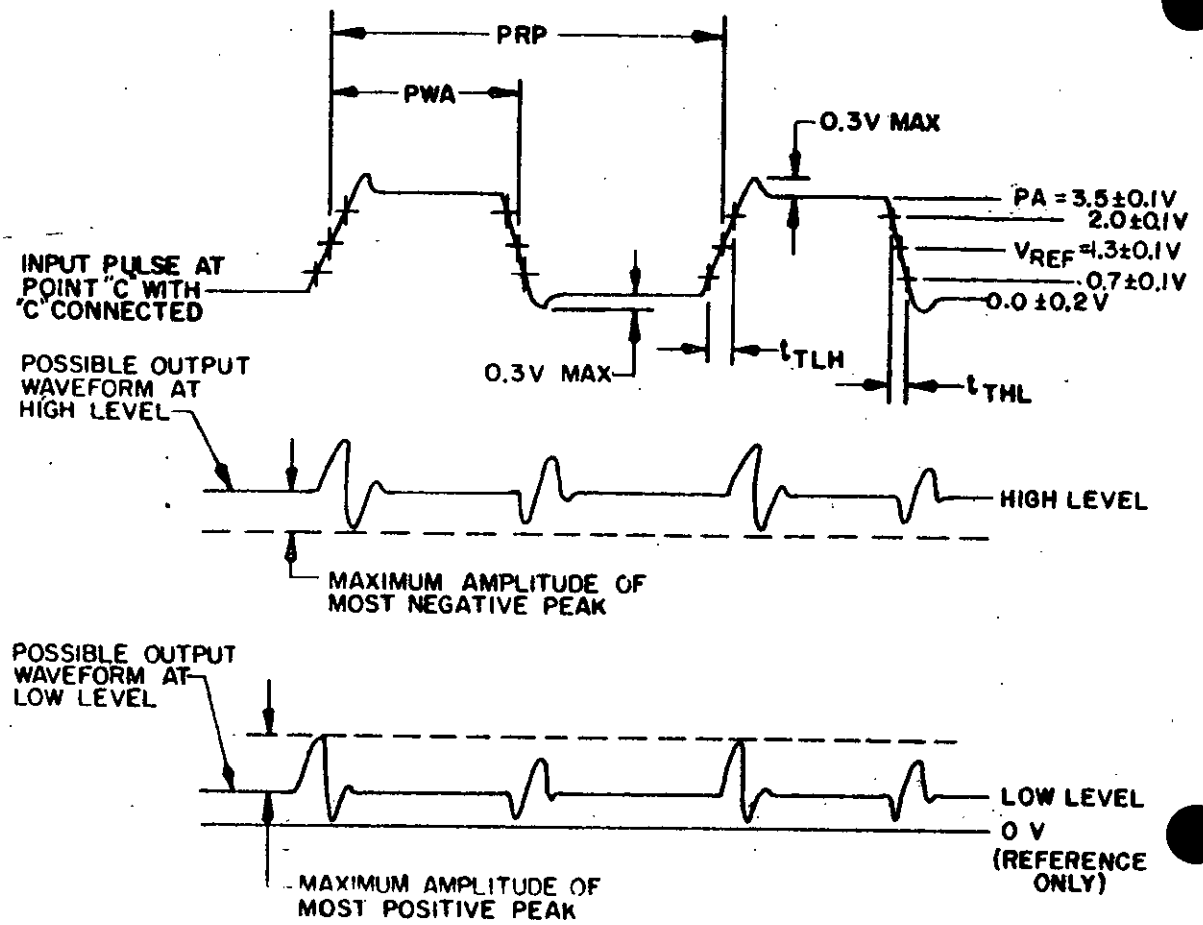
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Load all outputs with the load as specified on figure 9.

FIGURE 14. Output crosstalk (low-level and high-level) test circuit for tables XVI and XVII.

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NOTES:
Input pulse parameters:
$t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (verified at all inputs).
PWA = 200 ±20 ns.
PRP = 1,000 ±100 ns.

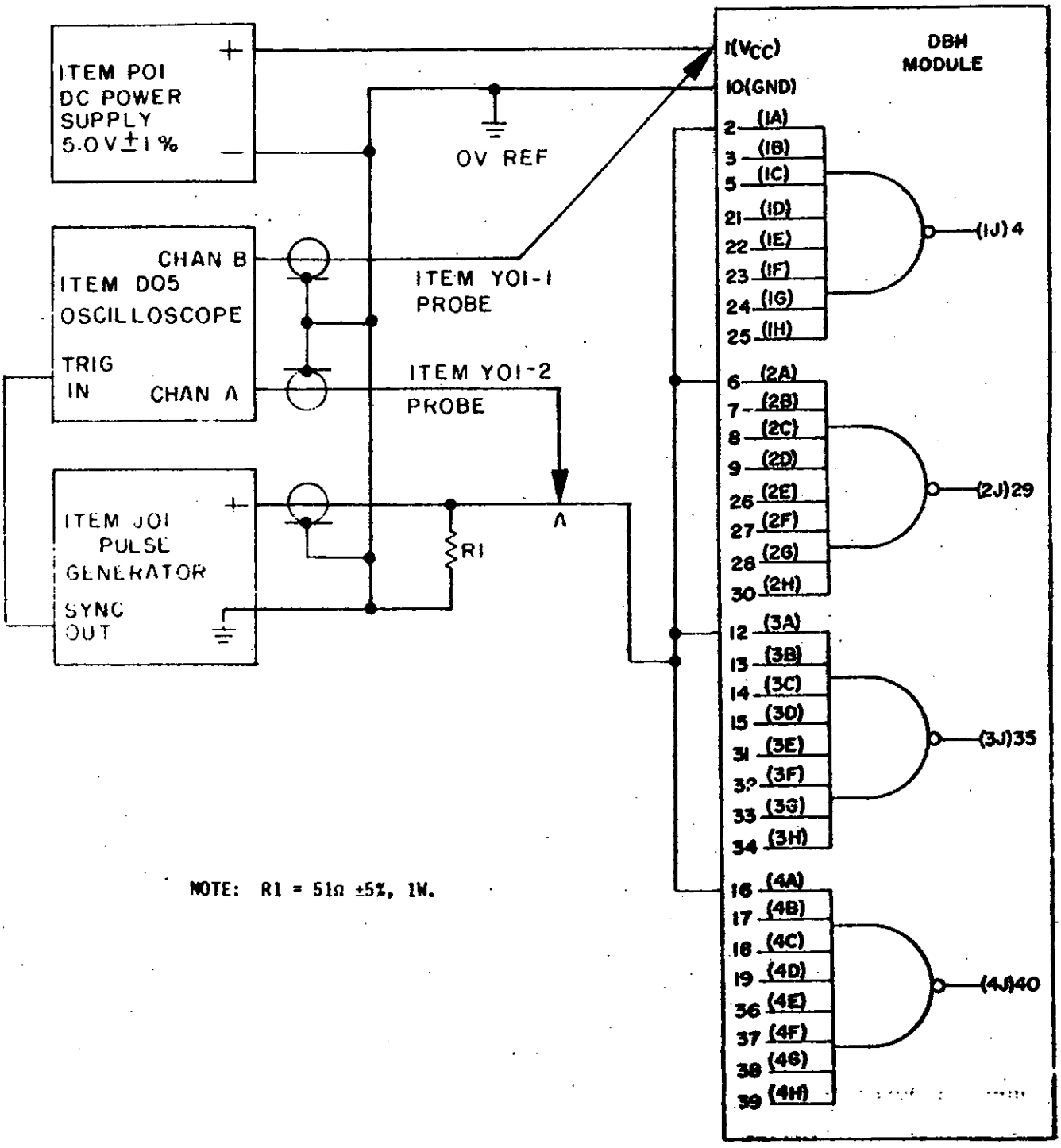
FIGURE 15. Output crosstalk (low-level and high-level) test circuit waveforms for tables XVI and XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
S	Power supply transient amplitude	0	300	mV	16, 17

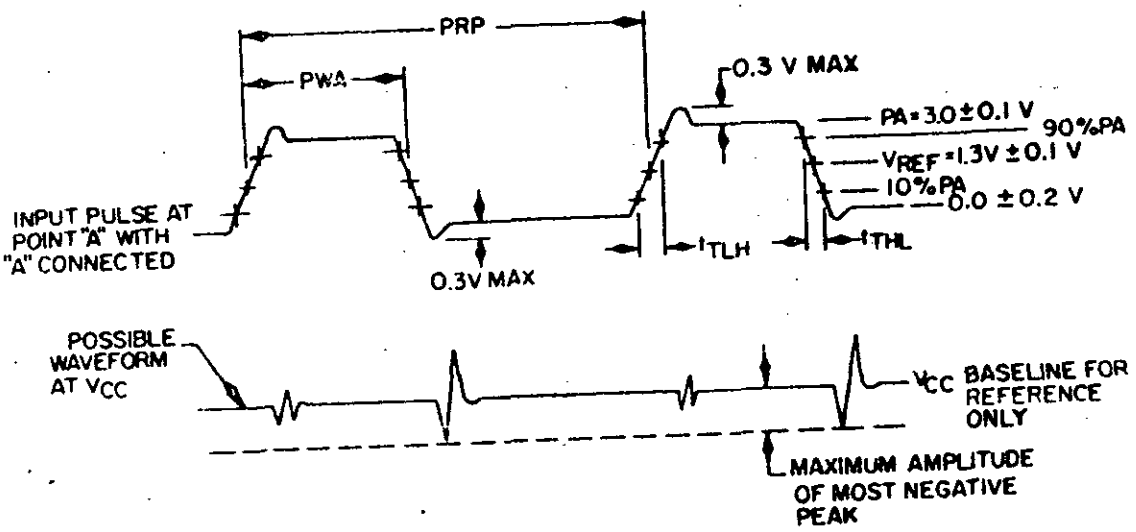
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:21Z
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NOTE: R1 = 51Ω ±5%, 1W.

FIGURE 16. Power supply transient amplitude test circuit for table XVIII.

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NOTES:
Input pulse parameters:
$t_{TLH} = t_{THL} = 20 \pm 2$ ns.
PWA = 200 ±20 ns.
PRP = 1,000 ns ±100 ns.

FIGURE 17. Power supply transient amplitude waveforms for table XVIII.

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TABLE XIX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T	Power supply filter capacitance	Open	1	1.65	2.75	µF	18

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TABLE XX. Input capacitance test.

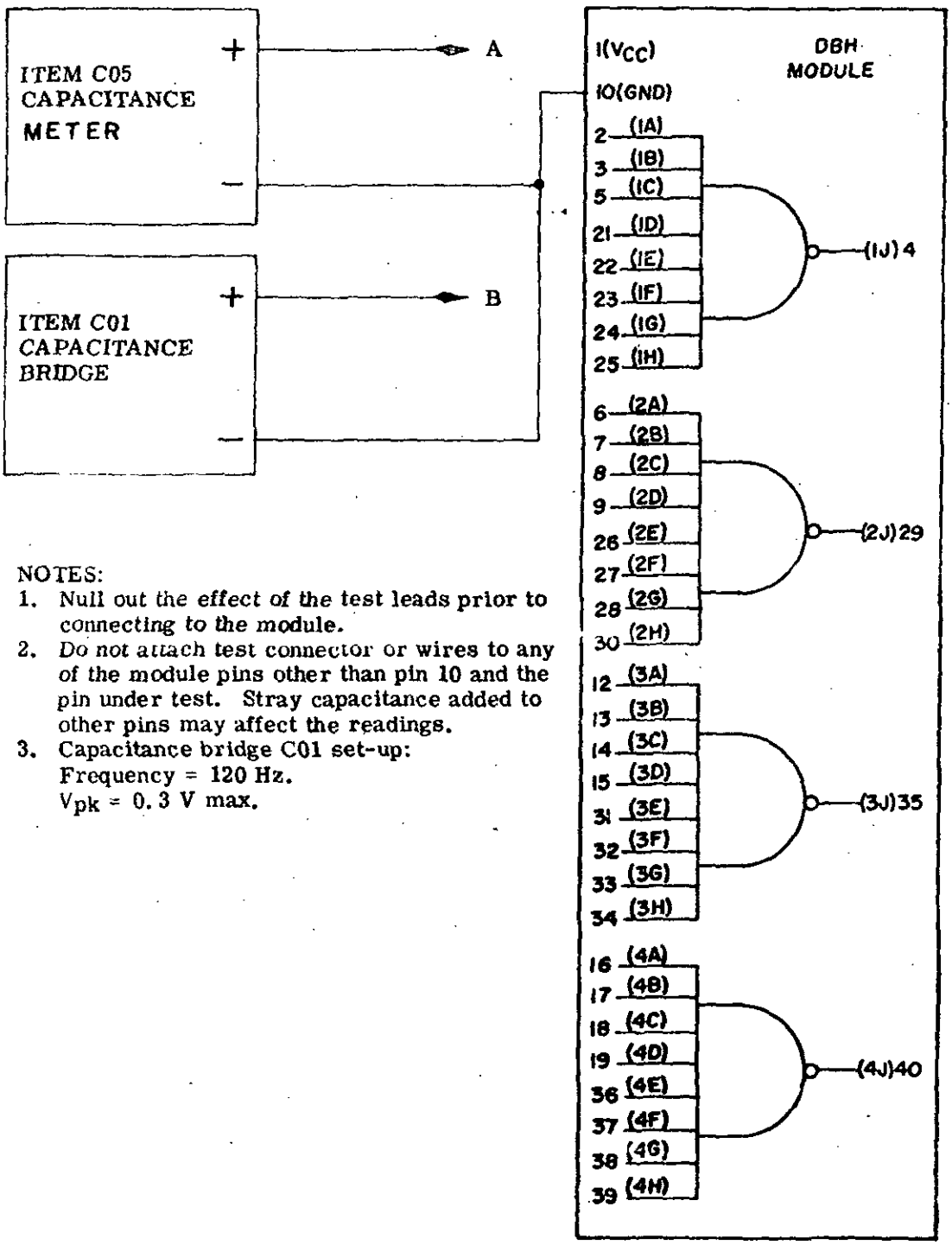
Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
U1	1A Input capacitance	2	2	15	pF	18
U2	1B	3				
U3	1C	5				
U4	1D	21				
U5	1E	22				
U6	1F	23				
U7	1G	24				
U8	1H	25				
U9	2A	6				
U10	2B	7				
U11	2C	8				
U12	2D	9				
U13	2F	27				
U14	2E	26				
U15	2G	28				
U16	2H	30				
U17	3A	12				
U18	3B	13				
U19	3C	14				
U20	3D	15				
U21	3E	31				
U22	3F	32				
U23	3G	33				

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TABLE XX. Input capacitance test - Continued.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
U24	3H Input capacitance	34	2	15	pF	18
U25	4A	16				
U26	4B	17				
U27	4C	18				
U28	4D	19				
U29	4E	36				
U30	4F	37				
U31	4G	38				
U32	4H	39				

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge C01 set-up:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V max.

FIGURE 18. Power supply filter capacitance and input capacitance test circuit for tables XIX and XX.

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